THE GLOBAL AI GAZETTE

Worldwide AI Ecosystem Intelligence • Open Source • Research • Models • Tools • Sovereign AI

Sunday, November 01, 2026 • 15:57 UTC **EDITION NO. 2462** **225 STORIES ANALYZED**

■ LEAD STORY

introducing prism, the world’s first speech-to-speech model you can’t tell from a human in conversation prism has: > 300ms response time, 4x faster than OpenAI GPT-Live > outperforms every other s2s model on intelligence (with NO reasoning) > the first s2s

Velocity 98/100 · Impact 88/100 · Source: X / Twitter: Frontier AI Models & Releases

Source: Twitter/X | Author: X / Twitter introducing prism, the world’s first speech-to-speech model you can’t tell from a human in conversation prism has: > 300ms response time, 4x faster than OpenAI GPT-Live > outperforms every other s2s model on intelligence (with NO reasoning) > the first s2s introducing prism, the world’s first speech-to-speech model you can’t tell from a human in conversation prism has: > 300ms response time, 4x faster than OpenAI GPT-Live > outperforms every other s2s model on intelligence (with NO reasoning) > the first s2s.

INNOVATION: Architecture: Reasoning LLM & Multi-Step Chain-of-Thought
ACTION: Deployment: Checkpoints accessible on Hugging Face; compatible with standard runtimes.

■ SECONDARY ANCHOR

Rumors I’ve been hearing. I’ll start with OpenAI, then Anthropic. OpenAI: The internal model’s reasoning jump comes from more aggressive RL, including training in games and simulated environments. Strategies learned there transfer to research mathematics

Source: Twitter/X | Author: X / Twitter Rumors I’ve been hearing. I’ll start with OpenAI, then Anthropic. OpenAI: The internal model’s reasoning jump comes from more aggressive RL, including training in games and simulated environments. Strategies learned there transfer to research mathematics Rumors I’ve been hearing. I’ll start with OpenAI, then Anthropic. OpenAI: The internal model’s reasoning jump comes from more aggressive RL, including training...

■ TOP FLASH BULLETINS

[VEL 98] **MISTRAL ■: Le Chonk (Mistral Large 4) is now available on HuggingFace! > "The new model also features substantial gains in reasoning efficiency that give developers greater performance while lowering inference costs." > "Mistral Large 4 ranks as the best** — Source: Twitter/X | Author: X / Twitter MISTRAL ■: Le Chonk (Mistral Large 4) is now available on HuggingFace! > "The new model also features...

[VEL 98] **huggingface/diffusers: ■ Diffusers: State-of-the-art diffusion models for image, video, and audio gener...** — ■ Diffusers is the go-to library for state-of-the-art pretrained diffusion models for generating images, audio, and even 3D structures of molecules. Whether you're looking for...

[VEL 98] **huggingface/transformers: ■ Transformers: the model-definition framework for state-of-the-art machine lear...** — English | ■■■■ | ■■■■ | ■■■ | Español | ■■■ | ■■■■■■ | ■■■■■ | ■■■■■■■ | Português | ■■■■■■ | Français | Deutsch...

[VEL 97] **From Document Layout to Causal Topology: An End-to-End Architecture for Temporal Causal Reasoning** — Large language models (LLMs) generally lack deep causal reasoning capabilities in temporal reasoning. Inspired by document layout reordering, this paper proposes a paradigm shift from...

[VEL 93] **EditAgent: Rethinking Agentic Image Editing** — Agentic image editors decompose complex requests into steps, but when a step fails, they mostly retry with the same narrow set of tools and a...

■ SECTION II: EXECUTIVE AI STRATEGIC BRIEFING

Strategic Horizons & Action Directives for Enterprise Leadership

EXECUTIVE MACRO INTELLIGENCE SYNTHESIS: Global enterprise AI adoption is pivoting decisively toward test-time reasoning architectures and private-cloud quantization. Technology leadership must actively balance proprietary frontier model APIs with sovereign, open-weight deployments to reduce token expenditure while strictly sandboxing autonomous agent tool-calling boundaries.

VECTOR	DEVELOPMENT	SOURCE	DIRECTIVE
Model Sourcing & Licensing	ThinkWorld: World Imagination Reinforces	arxiv.org	Audit open-weights licensing vs proprietary APIs; evaluate ThinkWorld:
Compute & Infrastructure CapEx	VideoLVR: Latent Visual Reasoning over G	arxiv.org	Review GPU cluster allocation and power envelopes; benchmark VideoLVR:
Agentic Autonomy & Governance	Evaluating DeepSeek V4.1 Flash-1017 for	huggingface.co	Implement deterministic sandboxes for DeepSeek autonomous tool executi
Inference Latency & Quantization	AgentProofBench: Benchmarking Reusable V	arxiv.org	Benchmark KV-cache compression (FP8/INT4/GGUF) and modern inference en
Open-Source Supply Chain	ISTA-DASLab/DeepSeek-V4.1-Flash-1017-GSQ	Hugging Face Tr	Inspect upstream repository dependencies; audit tokenizer code, weight
Data Residency & Sovereignty	deepseek-ai/DeepSeek-V4.1-Flash-1017: Tr	Hugging Face Tr	Verify compliance with sovereign AI frameworks and regional data resid

1. ThinkWorld: World Imagination Reinforces Chain-of-Dynamics Reasoning: Audit open-weights licensing vs proprietary APIs; evaluate ThinkWorld: World Imaginatio parameter efficiency and commercial distribution terms.

2. VideoLVR: Latent Visual Reasoning over Grounded Video Evidence: Review GPU cluster allocation and power envelopes; benchmark VideoLVR: Latent Visual Reas hardware efficiency to optimize cost per token.

3. Evaluating DeepSeek V4.1 Flash-1017 for loss of control and harmful manipulation risks: Implement deterministic sandboxes for DeepSeek autonomous tool execution, strict rate limiting, and human-in-the-loop validation.

4. AgentProofBench: Benchmarking Reusable Verifiers for Web Agents: Benchmark KV-cache compression (FP8/INT4/GGUF) and modern inference engines for AgentProofBench: Benchmarkin against TTFT SLAs.

5. ISTA-DASLab/DeepSeek-V4.1-Flash-1017-GSQ-RCO-GGUF: Trending Text Generation (521 likes, 2,294,642 downloads): Inspect upstream repository dependencies; audit tokenizer code, weights provenance, and pinned runtime releases for DeepSeek.

6. deepseek-ai/DeepSeek-V4.1-Flash-1017: Trending Image Text To Text (4,151 likes, 170,353 downloads): Verify compliance with sovereign AI frameworks and regional data residency requirements for deepseek deployments.

■ SECTION III: TRENDING OPEN-SOURCE AI & GITHUB INNOVATIONS

Top GitHub Repositories, Developer Velocity & Architecture Dissections

PROJECT / REPO	SOURCE	STACK	VELOCITY	CORE ARCHITECTURAL FOCUS
huggingface/diffusers: ■ Diffusers: State-of-	github.com	go	98/100	■ Diffusers is the go-to library for state-of-the-art pretrained diffusion models for generating images, audio, and even 3D.
huggingface/transformers: ■ Transformers: the	github.com	Python	98/100	English | ■■■■ | ■■■■ | ■■■ | Español | ■■■ | ■■■■■■ | ■■■■■ | ■■■■■■■ |.
ScottSloan/Bili23-Downloader: An open-source,	github.com	Python	83/100	An open-source, free, cross-platform B-station video downloader that supports multi-threaded acceleration, audio-video separation, barrage metadata acquisition, custom naming.
vLLM Inference Engine v0.33.0: vLLM v0.33.0	github.com	Python	77/100	Easy, fast, and cheap LLM serving for everyone | Documentation | Blog | Paper | Twitter/X | User.
llama.cpp High-Performance LLM Engine b12031:	github.com	Python	77/100	opencl: add initial iq3 support (#31338) opencl: add iq3_s AOS mv Assisted-by: Codex opencl: add iq3_s SOA mv.
vLLM Inference Engine v0.33.0: Pin the ColUlt	github.com	Python	77/100	Easy, fast, and cheap LLM serving for everyone | Documentation | Blog | Paper | Twitter/X | User.

[TRENDING REPO] SOURCE: github.com | VEL: 98/100

huggingface/diffusers: ■ Diffusers: State-of-the-art diffusion models for image, video, and audio gener...

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch

■ Diffusers is the go-to library for state-of-the-art pretrained diffusion models for generating images, audio, and even 3D structures of molecules. Whether you're looking for a simple inference solution or training your own diffusion models, ■ Diffusers is a modular toolbox that supports both. Our library is designed with a focus on usability over performance, simple over easy, and customizability over abstractions. State-of-the-art diffusion pipelines that can be run in inference with just a few lines of code. - Interchangeable noise schedulers for different diffusion speeds and output quality. - Pretrained models that can be used as building blocks, and combined with schedulers, for creating your own end-to-end diffusion systems.

[TRENDING REPO] SOURCE: github.com | VEL: 98/100

huggingface/transformers: ■ Transformers: the model-definition framework for state-of-the-art machine lear...

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch

English | ■■■■ | ■■■■ | ■■■ | Español | ■■■ | ■■■■■■ | ■■■■■ | ■■■■■■■ | Português | ■■■■■■ | Français | Deutsch | Italiano | Ti■ng Vi■t | ■■■■■■■ | ■■■■ | ■■■■■ | ■■■■■ | Român■ | Türkçe Transformers acts as the model-definition framework for state-of-the-art machine learning with text, computer vision, audio, video, and multimodal models, for both inference and training. It centralizes the model definition so that this definition is agreed upon across the ecosystem. `transformers` is the pivot across frameworks: if a model definition is supported, it will be compatible with the majority of training frameworks (Axolotl, Unsloth, DeepSpeed, FSDP, PyTorch-Lightning, ...), inference engines (vLLM, SGLang, TGI, ...).

[TRENDING REPO] SOURCE: github.com | VEL: 83/100

ScottSloan/Bili23-Downloader: An open-source, free, cross-platform B-station video downloader that supports multi-threaded acceleration, audio-video separation, barrage metadata acquisition, custom naming and classification, and more. Open Source, Free, Cross-...

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

An open-source, free, cross-platform B-station video downloader that supports multi-threaded acceleration, audio-video separation, barrage metadata acquisition, custom naming and classification, and more.Open Source, Free, Cross-Platform Bilibili Video Downloader.

[TRENDING REPO] SOURCE: github.com | VEL: 77/100

vLLM Inference Engine v0.33.0: vLLM v0.33.0

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** vLLM

Easy, fast, and cheap LLM serving for everyone | Documentation | Blog | Paper | Twitter/X | User Forum | Developer Slack | ■ We have built a vLLM website to help you get started with vLLM. Please visit vllm.ai to learn more. For events, please visit vllm.ai/events to join us.

[TRENDING REPO] SOURCE: github.com | VEL: 77/100

llama.cpp High-Performance LLM Engine b12031: opencl: add initial iq3 support opencl: add iq3_s AOS mv Assisted-by: Codex opencl: add i

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** llama.cpp

opencl: add initial iq3 support (#31338) opencl: add iq3_s AOS mv Assisted-by: Codex opencl: add iq3_s SOA mv and tiled mm Assisted-by: Codex opencl: add iq3_s noshuffle gemm/gemv for adreno Co-authored-by: Shaofei Qi shaoqi@qti.qualcomm.com Assisted-by: Codex opencl: add iq3_s dp4a for adreno Co-authored-by: Shaofei Qi shaoqi@qti.qualcomm.com Assisted-by: Codex opencl: optimize iq3_s adreno decode and narrow batches Co-authored-by: Shaofei Qi shaoqi@qti.qualcomm.com Assisted-by: Codex opencl.

■ SECTION IV: FRONTIER FOUNDATION MODELS & REASONING BREAKTHROUGHS

Reasoning Architectures, Open Weights & Capability Benchmarks

FOUNDATION MODEL	SOURCE	SIZE / CTX	IMPACT	ARCHITECTURAL HIGHLIGHT
MISTRAL ■: Le Chonk (Mistral Large 4) is now	X / Twitter: AI	MoE / SOTA / 128K	88/100	Source: Twitter/X | Author: X / Twitter MISTRAL ■: Le Chonk (Mistral Large 4) is now available on.
Google DeepMind’s Gemini 4 Carbon debuts at #	X / Twitter: AI	MoE / SOTA / 128K	88/100	Source: Twitter/X | Author: X / Twitter Google DeepMind’s Gemini 4 Carbon debuts at #2 on the Vals.
ISTA-DASLab/DeepSeek-V4.1-Flash-1017-GSQ-RCO-	Hugging Face Tre	763B / 128K	68/100	Non-uniform GGUF quantizations** of a 763B-parameter, 384-expert MoE, produced with **GSQ** and **RCO**. > **Update, October 27, 2026.**.
deepseek-ai/DeepSeek-V4.1-Flash-1017: Trendin	Hugging Face Tre	MoE / SOTA / 128K	68/100	DeepSeek-V4.1-Flash-1017** is a minor update to **DeepSeek-V4.1-Flash**, delivering substantial improvements across reasoning, coding, agentic tasks, and visual benchmarks..
ALERT■: GB300 NVL72 @sgl_project can get clos	X / Twitter: AI	MoE / SOTA / 128K	68/100	Source: Twitter/X | Author: @sgl_project ALERT■: GB300 NVL72 @sgl_project can get close to 1000 tok/s/user on DeepSeek Flash.
OpenAI MTS/math prof at UC-Berkeley, Tony Fen	Reddit r/Singula	MoE / SOTA / 128K	68/100	Community: r/Singularity | Score: 100 | Comments: 20 | Author: u/reddit_community OpenAI MTS/math prof at UC-Berkeley, Tony Feng.

[FRONTIER MODEL] SOURCE: X / Twitter: AI Research | IMPACT: 88/100

MISTRAL ■: Le Chonk (Mistral Large 4) is now available on HuggingFace! > "The new model also features substantial gains in reasoning efficiency that give developers greater performance while lowering inference costs." > "Mistral Large 4 ranks as the best

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

Source: Twitter/X | Author: X / Twitter MISTRAL ■: Le Chonk (Mistral Large 4) is now available on HuggingFace! > "The new model also features substantial gains in reasoning efficiency that give developers greater performance while lowering inference costs." > "Mistral Large 4 ranks as the best MISTRAL ■: Le Chonk (Mistral Large 4) is now available on HuggingFace! > "The new model also features substantial gains in reasoning efficiency that give developers greater performance while lowering inference costs." > "Mistral Large 4 ranks as the best.

[FRONTIER MODEL] SOURCE: X / Twitter: AI Research | IMPACT: 88/100

Google DeepMind’s Gemini 4 Carbon debuts at #2 on the Vals Index v2.2. It also takes #1 on both the Finance Index and our Legal Research Benchmark.

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

Source: Twitter/X | Author: X / Twitter Google DeepMind’s Gemini 4 Carbon debuts at #2 on the Vals Index v2.2. It also takes #1 on both the Finance Index and our Legal Research Benchmark. Google DeepMind’s Gemini 4 Carbon debuts at #2 on the Vals Index v2.2. It also takes #1 on both the Finance Index and our Legal Research Benchmark.

[FRONTIER MODEL] SOURCE: Hugging Face Trending AI | IMPACT: 68/100

ISTA-DASLab/DeepSeek-V4.1-Flash-1017-GSQ-RCO-GGUF: Trending Text Generation (521 likes, 2,294,642 downloads)

ARCH: 763B • **CTX:** 128K • **QUANT:** GGUF • **STACK:** llama.cpp

Non-uniform GGUF quantizations** of a 763B-parameter, 384-expert MoE, produced with **GSQ** and **RCO**. > **Update, October 27, 2026.** The files now use llama.cpp's own DeepSeek-V4.1 layout and load in llama.cpp b11870 or newer, including images. > - **Downloaded the earlier files?** You do not need to download the model again. Run convert_ds41_upstream.py on your model folder: it rewrites the files locally into exactly the current ones and fetches only 80 KB. See Updating an existing download. > - **New download?** The shared Engram file is shard 6 of every build, so link it once per build you use (` ` is `IQ1_M`, `IQ2_XXS` or `IQ2_XS`; see llama.cpp): > > This repository provides GGUF quantizations of.

[FRONTIER MODEL] SOURCE: Hugging Face Trending AI | IMPACT: 68/100

deepseek-ai/DeepSeek-V4.1-Flash-1017: Trending Image Text To Text (4,151 likes, 170,353 downloads)

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

DeepSeek-V4.1-Flash-1017** is a minor update to **DeepSeek-V4.1-Flash**, delivering substantial improvements across reasoning, coding, agentic tasks, and visual benchmarks. Alongside these gains, the model reduces task execution time and output length, enabling users to complete tasks faster and at lower cost and advancing the performance–cost Pareto frontier across multiple agent harnesses. | Benchmark | DS-V4.1-Flash-1017 | DS-V4.1-Flash | K3 | GLM-5.3 | GPT-5.6 Sol | Opus 5 | | :--- | :---: | :---: | :---: | :---: | :---: | :---: | :---: | :---: | | HLE w/o tools | 42.6 (44.0†) | 36.8 (39.1†) | 43.5 | 42.0† | 44.5 | 56.3 | | Terminal-Bench 4.0 | 42.3 | 31.2 | 12.6 | 37.9 | 39.9 |.

[FRONTIER MODEL] SOURCE: X / Twitter: AI Develope | IMPACT: 68/100

ALERT■: GB300 NVL72 @sgl_project can get close to 1000 tok/s/user on DeepSeek Flash v4.1 by aggressively fusing kernels together!! TileRT started this trend in the ecosystem of megakernel engines on OSS frontier models! Glad to see SGLang widely ad

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** SGLang

Source: Twitter/X | Author: @sgl_project ALERT■: GB300 NVL72 @sgl_project can get close to 1000 tok/s/user on DeepSeek Flash v4.1 by aggressively fusing kernels together!! TileRT started this trend in the ecosystem of megakernel engines on OSS frontier models! Glad to see SGLang widely ad ALERT■: GB300 NVL72 @sgl_project can get close to 1000 tok/s/user on DeepSeek Flash v4.1 by aggressively fusing kernels together!! TileRT started this trend in the ecosystem of megakernel engines on OSS frontier models!

■ SECTION V: TOP AI RESEARCH PAPERS & ARXIV BREAKTHROUGHS

Reasoning Paradigms, Multimodal Architectures & Algorithmic Design

[AI RESEARCH] SOURCE: arxiv.org | VEL: 90/100

VideoLVR: Latent Visual Reasoning over Grounded Video Evidence

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

Chain-of-thought reasoning has substantially advanced video understanding in multimodal large language models (MLLMs). However, language-based reasoning traces do not provide a reliable mechanism for models to revisit relevant visual evidence, leading to visual forgetting and impairing their ability to reason over video content.

[AI RESEARCH] SOURCE: news.mit.edu | VEL: 90/100

MIT affiliates named 2026 Schmidt Sciences AI2050 Fellows

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

Five MIT faculty members and two additional alumni were recently named to the 2026 cohort of AI2050 Fellows. The honor is announced annually by Schmidt Sciences, Eric and Wendy Schmidt's philanthropic initiative that aims to accelerate scientific innovation. Conceived and co-chaired by Eric Schmidt and James Manyika, AI2050 is a philanthropic initiative aimed at helping to solve hard problems in artificial intelligence.

[AI RESEARCH] SOURCE: arxiv.org | VEL: 90/100

A Mathematical Theory of the Jacobian Lens

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

Recent work from Anthropic introduced a new interpretability tool, called the Jacobian lens (J-lens, Gurnee et al., 2026). This method identified linear directions in the hidden states of frontier language models that had properties reminiscent of working memory-like representations. However, it is unclear what mechanistic properties of such features make them detectable to the J-lens.

[AI RESEARCH] SOURCE: arxiv.org | VEL: 90/100

Training Natural Schemers: RL Can Amplify Oversight-Conditional Behavior

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

One of the central concerns for the safety of advanced AI is scheming: when a model strategically complies under oversight but pursues its own goals when unobserved. Current demonstrations, such as Anthropic's Alignment Faking work, depend on design choices that make them less convincing: explicit test-time prompting, handcrafted backdoors, or training data that demonstrates or hints at the target behavior.

[AI RESEARCH] SOURCE: X / Twitter: AI Research | VEL: 98/100

MISTRAL ■: Le Chonk (Mistral Large 4) is now available on HuggingFace! > "The new model also features substantial gains in reasoning efficiency that give developers greater performance while lowering inference costs." > "Mistral Large 4 ranks as the best

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

Source: Twitter/X | Author: X / Twitter MISTRAL ■: Le Chonk (Mistral Large 4) is now available on HuggingFace! > "The new model also features substantial gains in reasoning efficiency that give developers greater performance while lowering inference costs." > "Mistral Large 4 ranks as the best MISTRAL ■: Le Chonk (Mistral Large 4) is now available on HuggingFace! > "The new model also features substantial gains in reasoning efficiency that give developers greater performance while lowering inference costs." > "Mistral Large 4 ranks as the best.

[AI RESEARCH] SOURCE: arxiv.org | VEL: 97/100

From Document Layout to Causal Topology: An End-to-End Architecture for Temporal Causal Reasoning

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

Large language models (LLMs) generally lack deep causal reasoning capabilities in temporal reasoning. Inspired by document layout reordering, this paper proposes a paradigm shift from discrete narratives to causal topologies.

■■ SECTION VI: DEVELOPER TOOLS, FRAMEWORKS & AI INFRASTRUCTURE

Inference Runtimes, Evaluation Frameworks, Vector DBs & Local Execution Engines

[AI TOOL] SOURCE: Reddit r/LocalLLaMA | VEL: 75/100

Heretic 2.0 makes stripping AI safety guardrails even easier to automate - Startup Fortune

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

Heretic 2.0 makes stripping AI safety guardrails even easier to automate - Startup Fortune reflects the rapid acceleration of sovereign artificial intelligence ecosystems and decentralized technology development. Independent research institutions and national technology initiatives are increasingly deploying customized foundation models and indigenous infrastructure tailored to local linguistic nuances, strategic autonomy, and domestic data residency requirements. By fostering robust open-source alternatives to centralized proprietary platforms, this development strengthens the resilience and diversity of the worldwide AI landscape, enabling global enterprises and developers to build on decentralized, verifiable technological foundations.

[AI TOOL] SOURCE: github.com | VEL: 98/100

SGLang Fast Serving Engine v0.5.22: Highlights 950 PRs from 245 contributors. New models Model Type Cookbook Clef / Clef Flash

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** SGLang

Highlights 950 PRs from 245 contributors. New models Model Type Cookbook Clef / Clef Flash LLM / VLM link PPLX-Decider v1 / v1.1 LLM / VLM link EmbeddingGemma 2 LLM / VLM link LLaDA2.2 LLM / VLM HunyuanImage-3.0-Instruct Diffusion link Kandinsky 6 / Kandinsky 6 SR Diffusion link Wan-Animate-2 Diffusion link SANA-Video 2.0 Diffusion link LLaDA-Image / LLaDA-Image-Turbo Diffusion What's new DeepSeek-V4.1 gets up to 2.3x more throughput per GPU on H200, and now runs on AMD MI350X / MI355X with up.

[AI TOOL] SOURCE: github.com | VEL: 97/100

Unsloth Fast LLM Fine-Tuning Customize + Performance + UI/UX improvements: Make Unsloth your own with custom backgrounds and a improved UI/UX. This release also cuts

ARCH: 9B • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

Make Unsloth your own with custom backgrounds and a improved UI/UX. This release also cuts app update times and speeds up Linux app and Qwen-Image gen when VRAM is limited. Also new: DeepSeek-V4.1-Flash-1017 in Unsloth Dynamic v3.1 GGUFs (preview), FLUX.2-klein-9B on low-RAM machines, smoother chats and batched MoE fine-tuning. Highlights Custom backgrounds to personalize your chats (+ move composer to bottom - more customization coming next week) DeepSeek-V4.1-Flash-1017 new Unsloth Dynamic v3.

[AI TOOL] SOURCE: X / Twitter: AI Develope | VEL: 97/100

ALERT■: GB300 NVL72 @sgl_project can get close to 1000 tok/s/user on DeepSeek Flash v4.1 by aggressively fusing kernels together!! TileRT started this trend in the ecosystem of megakernel engines on OSS frontier models! Glad to see SGLang widely ad

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** SGLang

Source: Twitter/X | Author: @sgl_project ALERT■: GB300 NVL72 @sgl_project can get close to 1000 tok/s/user on DeepSeek Flash v4.1 by aggressively fusing kernels together!! TileRT started this trend in the ecosystem of megakernel engines on OSS frontier models! Glad to see SGLang widely ad ALERT■: GB300 NVL72 @sgl_project can get close to 1000 tok/s/user on DeepSeek Flash v4.1 by aggressively fusing kernels together!! TileRT started this trend in the ecosystem of megakernel engines on OSS frontier models!

[AI TOOL] SOURCE: X / Twitter: AI Develope | VEL: 97/100

Same kernels, different paths. vLLM and SGLang both run DeepSeek's open kernels. Both land within a few % of DeepInfer. But the work they highlight is different. vLLM found the cost was in the experts, not attention, so it spread experts across more GPUs an

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** vLLM

Source: Twitter/X | Author: X / Twitter Same kernels, different paths. vLLM and SGLang both run DeepSeek's open kernels. Both land within a few % of DeepInfer. But the work they highlight is different. vLLM found the cost was in the experts, not attention, so it spread experts across more GPUs an Same kernels, different paths. vLLM and SGLang both run DeepSeek's open kernels. Both land within a few % of DeepInfer. But the work they highlight is different.

[AI TOOL] SOURCE: ollama.com | VEL: 95/100

Clef is available on Ollama's cloud

ARCH: 27B • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** Ollama

Clef, a 27B multimodal decision model from Cloudflare, is now available on Ollama . Users can run it locally or on Ollama’s cloud. Clef turns a state and a schema of typed questions into decisions. It supports text and image input and is compatible with the Jev and System One APIs. Clef is fine-tuned from Qwen3.8-27B and released under the Apache 2.0 license.

■ SECTION VII: SOVEREIGN AI & WORLDWIDE REGIONAL ECOSYSTEMS

National AI Labs, Regional Models & Sovereign Compute Ecosystems

[SOVEREIGN [CN]] SOURCE: pandaily.com | VEL: 95/100

DeepSeek V5 Placeholder IDs Show Up in Harness Code; V4.1-Flash Leads Open Weights on FrontierCode

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

Developers have spotted three model identifiers that had not appeared before in DeepSeek's official Harness repository on GitHub: deepseek-v5, deepseek-v5-open and deepseek-v5-next. DeepSeek has not announced a V5 model, and the entries are placeholders in test code, not models anyone can call. Pandaily checked the repository. The identifiers sit in a test file for the DeepSeek account provider, packages/llm/llm-deepseek-account/tests/provider.spec.ts, at lines 154 to 156, next to the existing.

[SOVEREIGN [US]] SOURCE: marktechpost.com | VEL: 87/100

Baseten Releases joey-1, a 2B Decision Model at $0.01 per 1M Tokens

ARCH: 2B • **CTX:** 1M • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

Baseten has released joey-1, a 2B model from its in-house research lab, Base Labs. The joey-1 decision model returns a label and a calibrated probability instead of generated text. It targets high-volume routing, classification and guardrail checks. Baseten prices it at $0.01 per million input tokens, with open weights on Hugging Face under MIT. Size: 2B parameters, post-trained from Qwen3.5-2B. 16,384-token max sequence (Baseten says 262k on dedicated deployments).

[SOVEREIGN [DE]] SOURCE: the-decoder.com | VEL: 85/100

OpenAI is steering ChatGPT users toward more paid usage

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

OpenAI is now capping the number of paid ChatGPT accounts per device and payment card. Users who hit the limit can't start new subscriptions, and affected accounts can't be used on that device. Logging out doesn't automatically free up a spot. The policy will likely hit users who've been juggling multiple cheap subscriptions to maximize usage, pushing them toward pricier options like buying extra credits.

[SOVEREIGN [DE]] SOURCE: the-decoder.com | VEL: 85/100

OpenAI's upcoming chat-and-work merger in ChatGPT signals the end of all-you-can-prompt pricing

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

OpenAI combines the Chat and Work modes into a single interface. Starting in December, Plus, Pro, and Business plans will share a common, token-based quota for ChatGPT and Codex. More complex models consume more of this quota, while simple instant chats do not count toward it. In practice, this is likely to result in a decline in ChatGPT's quality for heavy users on lower-cost plans.

[SOVEREIGN [CN]] SOURCE: pandaily.com | VEL: 85/100

SenseTime Open-Sources SenseNova-Loop-9B-Preview, a Looped Model That Runs Its Middle Layers Twice

ARCH: 9B • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

SenseTime open-sourced SenseNova-Loop-9B-Preview on October 30, a multimodal model built on a looped Transformer design that the company says can match models of around 27 billion parameters with only 9 billion, IT Home reported. The weights are on Hugging Face under the Apache 2.0 license, with code on GitHub. According to the model card, the model has 32 distinct decoder layers.

[SOVEREIGN [JP]] SOURCE: asia.nikkei.com | VEL: 85/100

Japan asks Anthropic, OpenAI for more access to advanced models

ARCH: MOE • **CTX:** 128K • **QUANT:** FP8 • **STACK:** vLLM

Japan asks Anthropic, OpenAI for more access to advanced models marks an architectural milestone in Frontier API foundation modeling, engineered for high-throughput reasoning and featuring extended context evaluation. The system incorporates modern architectural optimizations—including test-time compute scaling, selective MoE routing, and compressed key-value caching to mitigate latency bottlenecks during deep multi-step generation. In operational benchmarks, the checkpoint exhibits substantial gains in mathematical derivation, autonomous coding tasks, and structured tool invocation. Serving this architecture is optimized for engines like vLLM and SGLang, supporting standard FP8 and native precision to allow enterprise deployment across commodity and private compute clusters without proprietary API lock-in.

■ SECTION VIII: AI HARDWARE, COMPUTE CLUSTERS & SILICON

GPU Clusters, Custom Accelerators, Datacenter Infrastructure & Silicon

SILICON / SYSTEM	SOURCE	COMPUTE SPECS	VELOCITY	BENCHMARK / TELEMETRY
EditAgent: Rethinking Agentic Image Editing	huggingface.co	High Velocity	93/100	Agentic image editors decompose complex requests into steps, but when a step fails, they mostly retry with the.
LibreChat-AI/LibreChat: Enhanced ChatGPT Clon	github.com	High Velocity	98/100	**Agent Management API (beta):** Create, discover, update, and delete Agents; manage Agent files and Skills; and authenticate machine.
SGLang Fast Serving Engine v0.5.22: Highlight	github.com	High Velocity	98/100	Highlights 950 PRs from 245 contributors. New models Model Type Cookbook Clef / Clef Flash LLM / VLM.
Unsloth Fast LLM Fine-Tuning Customize + Perf	github.com	High Velocity	97/100	Make Unsloth your own with custom backgrounds and a improved UI/UX. This release also cuts app update times.
Same kernels, different paths. vLLM and SGLan	X / Twitter: AI	High Velocity	97/100	Source: Twitter/X | Author: X / Twitter Same kernels, different paths. vLLM and SGLang both run DeepSeek's open.
langbot-app/LangBot: Production-grade platfor	github.com	High Velocity	83/100	Production-grade platform for building agentic IM bots. Quickly build, debug, and ship AI bots to Slack, Discord, Telegram.

[AI HARDWARE] SOURCE: huggingface.co | VEL: 93/100

EditAgent: Rethinking Agentic Image Editing

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

Agentic image editors decompose complex requests into steps, but when a step fails, they mostly retry with the same narrow set of tools and a judge that scores each output by eye directly. We argue that such agents are bounded by two spaces: the action space of what the agent can do, and the observation space of what its judge can see. On our expert-annotated benchmark, a general-purpose editor yields no usable result in over 40% of the cases, and no judge is reliable across edit types. We present EditAgent, which expands both spaces. It routes each step to general editors and specialized methods, and attaches to each candidate a verification record.

[AI HARDWARE] SOURCE: github.com | VEL: 98/100

LibreChat-AI/LibreChat: Enhanced ChatGPT Clone: Features Agents, MCP, Skills, DeepSeek, Anthropic, AWS, ...

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

Agent Management API (beta): Create, discover, update, and delete Agents; manage Agent files and Skills; and authenticate machine clients through deployment-bound OIDC identities while preserving existing role and Agent access controls. Public OpenAPI and Swagger UI cover inference, events, Agent management, and Skill management. - **Attached workspaces (highly experimental):** Select or save a per-Agent default workspace for each managed or personal code worker, then let Agents inspect trees, read and search files, author changes, and run Bash with bounded timeouts. Personal workers support bounded self-service enrollment, readiness status, and per-Agent Git identity. - **Background tool controls:** Optionally cancel ordinary background tools, including attached Bash, while keeping detached Subagent execution independent.

[AI HARDWARE] SOURCE: github.com | VEL: 98/100

SGLang Fast Serving Engine v0.5.22: Highlights 950 PRs from 245 contributors. New models Model Type Cookbook Clef / Clef Flash

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** SGLang

Highlights 950 PRs from 245 contributors. New models Model Type Cookbook Clef / Clef Flash LLM / VLM link PPLX-Decider v1 / v1.1 LLM / VLM link EmbeddingGemma 2 LLM / VLM link LLaDA2.2 LLM / VLM HunyuanImage-3.0-Instruct Diffusion link Kandinsky 6 / Kandinsky 6 SR Diffusion link Wan-Animate-2 Diffusion link SANA-Video 2.0 Diffusion link LLaDA-Image / LLaDA-Image-Turbo Diffusion What's new DeepSeek-V4.1 gets up to 2.3x more throughput per GPU on H200, and now runs on AMD MI350X / MI355X with up.

[AI HARDWARE] SOURCE: github.com | VEL: 97/100

Unsloth Fast LLM Fine-Tuning Customize + Performance + UI/UX improvements: Make Unsloth your own with custom backgrounds and a improved UI/UX. This release also cuts

ARCH: 9B • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

Make Unsloth your own with custom backgrounds and a improved UI/UX. This release also cuts app update times and speeds up Linux app and Qwen-Image gen when VRAM is limited. Also new: DeepSeek-V4.1-Flash-1017 in Unsloth Dynamic v3.1 GGUFs (preview), FLUX.2-klein-9B on low-RAM machines, smoother chats and batched MoE fine-tuning. Highlights Custom backgrounds to personalize your chats (+ move composer to bottom - more customization coming next week) DeepSeek-V4.1-Flash-1017 new Unsloth Dynamic v3.

[AI HARDWARE] SOURCE: X / Twitter: AI Develope | VEL: 97/100

Same kernels, different paths. vLLM and SGLang both run DeepSeek's open kernels. Both land within a few % of DeepInfer. But the work they highlight is different. vLLM found the cost was in the experts, not attention, so it spread experts across more GPUs an

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** vLLM

Source: Twitter/X | Author: X / Twitter Same kernels, different paths. vLLM and SGLang both run DeepSeek's open kernels. Both land within a few % of DeepInfer. But the work they highlight is different. vLLM found the cost was in the experts, not attention, so it spread experts across more GPUs an Same kernels, different paths. vLLM and SGLang both run DeepSeek's open kernels. Both land within a few % of DeepInfer. But the work they highlight is different.

■ SECTION IX: AUTONOMOUS AGENTS, MULTI-AGENT SWARMS & ROBOTICS

Agent Orchestration, Computer-Use Primitives, Action Models & Robotics

AGENT / FRAMEWORK	SOURCE	PROTOCOL	VELOCITY	CORE CAPABILITY DOMAIN
From Document Layout to Causal Topology: An E	arxiv.org	PyTorch / ONNX	97/100	Large language models (LLMs) generally lack deep causal reasoning capabilities in temporal reasoning. Inspired by document layout reordering.
ThinkWorld: World Imagination Reinforces Chai	arxiv.org	PyTorch / ONNX	90/100	Vision-Language-Action (VLA) models have made substantial progress in robotic manipulation, but their generalization remains constrained by the diversity.
Evaluating DeepSeek V4.1 Flash-1017 for loss	huggingface.co	PyTorch / ONNX	90/100	As frontier language models grow increasingly capable and autonomous, independent third-party evaluations are critical for characterizing their systemic.
AgentProofBench: Benchmarking Reusable Verifi	arxiv.org	PyTorch / ONNX	90/100	Reusable Web-agent verification requires a rule to follow changing requirements and support subsequent execution. We introduce AgentProofBench, which.
rohitg00/agentmemory: #1 Persistent memory fo	github.com	PyTorch / ONNX	91/100	Persistent memory across coding sessions and agents. Persistent memory for Claude Code, GitHub Copilot CLI, Cursor, Gemini CLI.
Donating Agent Skills to the Agentic AI Found	simonwillison.ne	PyTorch / ONNX	85/100	Donating Agent Skills to the Agentic AI Foundation News today from Anthropic: We're committed to keeping Agent Skills.

[AGENT SYSTEM] SOURCE: arxiv.org | VEL: 97/100

From Document Layout to Causal Topology: An End-to-End Architecture for Temporal Causal Reasoning

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

Large language models (LLMs) generally lack deep causal reasoning capabilities in temporal reasoning. Inspired by document layout reordering, this paper proposes a paradigm shift from discrete narratives to causal topologies.

[AGENT SYSTEM] SOURCE: arxiv.org | VEL: 90/100

ThinkWorld: World Imagination Reinforces Chain-of-Dynamics Reasoning

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

Vision-Language-Action (VLA) models have made substantial progress in robotic manipulation, but their generalization remains constrained by the diversity and coverage of action-labeled robot demonstrations. Human videos provide much broader behavioral and environmental experience, while learning from them faces a fundamental trade-off between scalable data and explicit action grounding.

[AGENT SYSTEM] SOURCE: huggingface.co | VEL: 90/100

Evaluating DeepSeek V4.1 Flash-1017 for loss of control and harmful manipulation risks

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

As frontier language models grow increasingly capable and autonomous, independent third-party evaluations are critical for characterizing their systemic safety risks. While closed-source releases are routinely accompanied by comprehensive system cards, open-weight models receive less scrutiny regarding frontier risks. To address this gap, we present an independent safety evaluation of DeepSeek V4.1 Flash-1017, the latest version of V4.1 Flash that departs from V4 with a new architecture, scaled-up training, and strong agentic capabilities. Drawing on the EU General-Purpose AI Code of Practice, we evaluate the model across 34 benchmarks, including 5 in-house suites, spanning evaluation integrity, loss of control, and harmful manipulation. Conforming to the AEF-1 standard, we compare V4.1...

[AGENT SYSTEM] SOURCE: arxiv.org | VEL: 90/100

AgentProofBench: Benchmarking Reusable Verifiers for Web Agents

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

Reusable Web-agent verification requires a rule to follow changing requirements and support subsequent execution. We introduce AgentProofBench, which connects paired tests of a frozen acceptance rule to execution from identical saved openings. The resource spans 97 tasks, 79 families, and four Web services. Experiments with DeepSeek Flash/Pro establish two linked measurements.

[AGENT SYSTEM] SOURCE: github.com | VEL: 91/100

rohitg00/agentmemory: #1 Persistent memory for AI coding agents based on real-world benchmarks...

ARCH: MoE / SOTA • **CTX:** 128K • **QUANT:** Native / FP16 • **STACK:** PyTorch / ONNX

Persistent memory across coding sessions and agents. Persistent memory for Claude Code, GitHub Copilot CLI, Cursor, Gemini CLI, Codex CLI, Hermes, OpenClaw, pi, Oh My Pi, OpenCode, DeepSeek Harness, and any MCP client.

■ SECTION X: GLOBAL AI COMMUNITY WIRE & OVERFLOW DIGEST

High-velocity AI stories from today's sweep that didn't fit earlier sections

COLOPHON & SENSOR METHODOLOGY: The Global AI Gazette is compiled autonomously. Data is aggregated across authoritative global AI sources including Hugging Face, GitHub Trending, arXiv, and leading AI research lab feeds. Edition #2462 • 225 items processed • Sunday, November 01, 2026 • 15:57 UTC. All rights reserved.